

Chemistry — Chapters 1 to 9

Revision Test 2026 | SSC-I (9th Grade) | Version 1 | 31-mark no-choice paper
 Candidate: Seemab | Date: 18 August 2026 | Total Marks: 31 | Time allowed: 1 h 30 m

29 / 31TOTAL SCORE
2 MCQs voided on SLO audit**93.5%**PERCENTAGE
first issued as 87.1%**A+**GRADE
was reported as A**12 / 12**MCQ
10 correct, 2 voided**Two questions voided on syllabus audit, and both marks awarded**

This paper was re-audited against the Student Learning Outcomes printed on page 1 of each chapter, the same check that was applied to Physics exam-13. **Chapter 4's SLO list contains no outcome covering either of these two questions**, so both fail the rule that a question needs explicit book content **and** a stated SLO.

Q5, who discovered the atomic number in 1913. The sentence is printed at Ch4 §4.1 p.46, but Chapter 4's outcomes ask only that she can "define the periodic table as an arrangement element in periods and group, in order of increasing proton number/atomic number". That is the **fact**, not the history of who found it or when. **Out of syllabus.**

Q6, the number of elements in period 6. Printed at Ch4 §4.1.1 p.47 and reinforced by Activity 4.1, but again **no SLO asks for the count of elements per period**. Seemab had scribbled this answer out and left it unreadable; the question is now void regardless of what she intended, so the open item is closed and the mark is hers. **Out of syllabus.**

Score: **27/31 (87.1%, A)** as first marked, now **29/31 (93.5%, Grade A+)**. The remaining ten MCQs were all answered correctly, so her Section A is **12/12**.

Section A — Multiple Choice Questions (12 × 1 = 12 Marks) — 12/12 after voids

#	Question (abbreviated)	Seemab	Correct	Result
1	Branch designing safer, less hazardous products	B	B (Green chemistry)	CORRECT
2	Liquid crystals are used in	B	B (LCD display screens)	CORRECT
3	Element existing in different physical forms	B	B (Allotropy)	CORRECT
4	Neutrons in ^{56}Fe ($Z = 26$)	B	B (30)	CORRECT
5	Who discovered atomic number in 1913	C	B (Moseley) — no Ch4 SLO covers the history	VOID, AWARDED
6	Number of elements in the sixth period	blot	D (32) — no Ch4 SLO covers elements per period	VOID, AWARDED
7	Each K atom in K_2O	B	B (loses 1 electron)	CORRECT
8	Empirical formula of ethene C_2H_4	B	B (CH_2)	CORRECT
9	Which substance uses "formula mass"	B	B (NaCl)	CORRECT
10	Coefficient of O_2 balancing C_2H_2 combustion	C	C (5)	CORRECT
11	Oxidation number of Cl in KClO_2	B	B (+3)	CORRECT
12	Enthalpy relationship in an exothermic reaction	B	B ($H_{\text{prod}} < H_{\text{react}}$)	CORRECT

MCQ score: **12/12**. She answered ten correctly, and the only two she did not (Q5 and Q6) have both been voided as out of syllabus. **Every question on this paper that was actually within her course, she got right.**

Section B — Short Questions (3 × 3 = 9 Marks) — 9/9

Q	Topic	Awarded	Max	Examiner's note
2(i)	Molar mass of SO_2 ; mass of one molecule	3	3	$M = 32 + 2(16) = 64$ correct. Then $64 \div 6.022 \times 10^{23} = \mathbf{1.06 \times 10^{-22} \text{ g}}$, fully correct. She first wrote "mass = molar mass $\times N_A$ " and self-corrected to division before solving, which is the right instinct. Full marks.
2(ii)	Dot-and-cross of H_2O , bond type, lone pairs	3	3	Diagram shows two shared pairs and two lone pairs on oxygen. Wrote "Covalent bonds" and "Oxygen has 2 lone pairs of electron." Both correct. Full marks.
2(iii)	Oxidation numbers in H_2SO_3 , Na_2CO_3 , KIO_4	3	3	$S = +4$, $C = +4$, $I = +7$. All three correct with working shown line by line. This was the hardest-to-guess question on the paper and she got every part.

Section C — Long Questions (2 × 5 = 10 Marks) — 8/10

Q	Topic	Awarded	Max	Examiner's note
3(i)	Empirical → molecular formula; moles in 18 g	4	5	(a) 3/3 — 100 g basis, moles 2.225 / 2.2 / 4.443, divided by the smallest to

				give 1 : 1 : 2 and CHO₂ . Textbook execution. (b) 1/1 — empirical mass 45, $90 \div 45 = 2$, C₂H₂O₄ . (c) 0/1 — wrote "molar mass of CHO ₂ :- 45 g" and computed $18 \div 45 = 0.4 \text{ mol}$. The question asks for moles of <i>the compound</i> , which is C ₂ H ₂ O ₄ with M = 90, so it is $18 \div 90 = 0.2 \text{ mol}$. She used the empirical unit mass instead of the real molar mass.
3(ii)	Silver isotopes; sub-atomic particles; nitrogen config	4	5	(a) 2/2 — $(107 \times 52 + 109 \times 48) \div 100 = 10796 \div 100 = 107.96$. Correct. (b) 1/1 — protons 47, electrons 47, neutrons $109 - 47 = 62$. Correct. (c) 1/2 — configuration 1s² 2s² 2p³ correct, but then she wrote "group :- 2 nd " and "period :- $2 + 3 + 10 = 13$ or 3". She has swapped the two concepts : the 2 belongs to the period, not the group. Correct answer is Group 15 (VA), Period 2.

What went well

Section B was flawless, and it was the most technically demanding part of the paper: a single-molecule mass calculation and three oxidation numbers on compounds she has never been given before (H₂SO₃, Na₂CO₃, KIO₄). The empirical-formula chain in 3(i)(a) and the silver weighted average in 3(ii)(a) were both carried out cleanly with correct method and correct arithmetic. Stoichiometry remains her strongest area in this subject.

Revision priorities from this paper

- **Empirical vs molecular molar mass.**

The one real Section C slip. Once the molecular formula is found, every later mass or mole calculation must use the *molecular* molar mass. Drill: "which mass do I divide by?" as an explicit step.

- **Group and period from electron configuration.**

She has these the wrong way round. Fix the rule:

number of valence electrons → group

;

number of occupied shells → period

. For nitrogen, 5 valence electrons means Group 15, and 2 shells means Period 2.

- **Unit slip worth naming even though no mark was lost:**

she wrote the molar mass as "64 mol/g" instead of g/mol, and again "0.4 mol/g" in 3(i)(c). The value was right so nothing was deducted, but the habit will cost marks in a stricter board paper.